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tijhuisingenieurs.nl/files/2019-02-landwater-meerjarig-onderzoek-naar-aanwas-van-bagger.pdf

Multi-year research into dredging sludge accretion

Water boards would like to have more accurate accretion figures so that dredging can be planned in a more risk-based way. The research project Inzicht in BaggerAanwas (IBA) takes a different direction from earlier studies, partly by classifying watercourses on the basis of environmental factors such as water width, the presence of trees and land use.

Over the past twenty years, a great deal of research has already been carried out to gain more insight into sludge accretion in watercourses. This ranges from projects with sludge traps, comparative studies of electronic depth measurements versus manual profile measurements (with or without an intelligent depth gauge), and scientific discussions on interpolation aimed at calculating average profile characteristics per watercourse. This research has been carried out both in the Netherlands and abroad.

IBA does things differently. At a large number of measurement locations, field workers measure profiles at exactly the same location in at least three consecutive measurement years. The result is then examined statistically, so that a statement can be expected to be made about dredging sludge accretion per type of watercourse. By classifying watercourses on the basis of environmental factors, the result can be extrapolated to other watercourses in a similar situation.

Collaboration

In 2016, seven water boards jointly decided to put the multi-year research out to tender. These are Hoogheemraadschap van Delfland, Hoogheemraadschap Hollands Noorderkwartier, Hoogheemraadschap van Rijnland, Hoogheemraadschap van Schieland en de Krimpenerwaard, Hoogheemraadschap de Stichtse Rijnlanden, Waterschap Rivierenland and Amstel Gooi en Vecht.

The contractor is a consortium of Niebeek Milieumanagement, Ingenieursbureau Land, VanderHelm and Delta Marking. Tijhuis Ingenieurs and Hussem Consultancy have been engaged to supervise the quality of the data collection and the data delivered. They are also responsible for project support and analyse the data.

A new type of approach

Spread across the area of the seven water boards, 135 watercourses (measurement locations) have been selected that may be distinguishing for dredging sludge accretion. Through different combinations of environmental factors, these represent 21 types of watercourse. For three types of watercourse, fifteen measurement locations per type have been selected, and for the remaining eighteen types of watercourse, five measurement locations. On average, twenty cross-sectional profile measurements have been placed at 10-metre intervals at each measurement location. For a large proportion of the profiles, the measuring interval within a profile is 50 cm. In total, almost 2,500 cross-sectional profiles are measured per measurement round.

The variation in bed level can generally differ greatly along the length of a watercourse. As a result, the profiles in subsequent years must be placed within one metre of the profile measurements of

the first measurement year. An extremely accurate XY position of the entire profile is therefore of great importance. The height reference (Z) must also be checked several times a day, including by means of fixed points, because RTK-GPS is known to vary during the day. These fixed points must not change during the measurement years of the project. The measurement procedure and associated records are more extensive than for regular depth-survey work. In collaboration with the client, a project-specific measurement protocol has been drawn up, in which the method of recording and the checks to be carried out are laid down. In the meantime, the experience gained regarding the method of measuring, recording and data processing has led to optimisation of the measurement protocol.

Reliable

Demonstrability and traceability of measurement data are crucial within the IBA project.

The bed level is measured at fixed locations in the classified watercourses.

IN BRIEF - Dredging

In this project, sludge accretion is linked to environmental factors

Seven water boards have put multi-year research out to tender

135 watercourses (measurement locations) have been selected across the area

Demonstrability and traceability of measurement data are crucial within the project

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This requires a great deal of effort from both the field staff and the data specialists of the contractors and the client. The field staff's experiences are positive. They find that the measuring is essentially no different, but that the interim checks and the recording of them are far more extensive than with regular water level surveys, which means it takes more time than usual. At the start of the project, this took some getting used to for the field staff, but they have handled it well. The measurement locations are sometimes far apart, which proved a planning challenge for efficient measuring.

Processing the collected data also takes more time than with 'normal' water level surveys. More is recorded and accuracy is of greater importance. Both in the field and in the office, 'a good and reliable set of measurement data' comes first. Good consultation between the contractors and the clients contributes to the high quality of the dataset. The constant improvement, the collaborative thinking as a project team with the client(s) and the cooperation with fellow companies are experienced as positive.

The environmental factors (characteristics) are assessed annually to gain an insight into any changes in these factors. These include the felling or pruning of trees bordering the watercourse or newly installed bank protection at the measurement location.

Electronic measuring

On fifteen of the 135 selected measurement sections, the water bed is also surveyed electronically. Using a 'single-beam' echo sounder, a boat designed especially for this project measures the top of the sludge layer. A minimum water depth of 40 cm is required for good and reliable measurements. The purpose of these measurements is to determine whether accretion can also be determined in this way. The same quality requirements as for manual measuring apply to this method, although they are assured in other ways. For example, there is measuring equipment on board that determines, among other things, the squat when the boat is propelled forward and thus corrects the measured depth values.

Challenges

To ensure readily comparable conditions, the timing of the measurements in successive years may not vary by more than one month (if a location was measured in January in measurement round 1, it must also be measured in January in the following years). In addition, the same survey team must measure the same profiles each year.

A lesson learned from the first measurement year is that wind force 4 and above can adversely affect the accuracy of the data. The environmental factors can also change in the interim due to interventions such as dredging. Around several measurement locations it has already been established that work has taken place, such as the installation of new bank protection or clearing, during which a little sludge was often removed as well.

The project group has spread the measurement sections across the participating water boards in order to limit non-routine work around the measurement sections as much as possible. In this way, the aim is to be able to carry out representative multi-year measurements.

Quality control

Both the contractor and the supervisor pay close attention to compliance with and interpretation of the measurement protocol and the required quality of the data collected. Before each measurement

year, the contractor organises a joint fieldwork day to standardise and refresh the measurement method and recording procedures of all fieldwork teams. The parties involved consult one another periodically. Among other things, unexpected matters from the field are discussed, after which the working method can be tightened up again without making the measurement protocol needlessly complicated.

The collecting party, the supervising party and the commissioning party are all fully aware of the importance of this multi-year study and the quality required for it. This increases the likelihood that after measurement round 3, a statement can be made about a specific sludge accretion rate for a set of characteristic types of watercourse that are common in the area of the seven water boards. We do not rule out an extension of the project. With an extension, the measurement series would be expanded, so that we expect to be able to determine the accretion rate more accurately.

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Seven water boards are working together on a project to study dredging sludge accretion over several years.

A special boat has been developed for this project that measures the top of the dredged sludge layer.

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